

Enterprise AI Evaluation Report

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Task Type : Clustering

Target Column: None

Best Model : DBSCAN

Best Score : 0.9949

The dataset contains 213,768 records across 2 features (1 numerical, 1 categorical). The dataset is complete – no missing values detected. | Pipeline: median imputation for continuous features (resistant to skew), mode imputation for categoricals. IQR-based outlier flagging applied. One-Hot Encoding via `pd.get_dummies` (drop first=True to avoid multicollinearity).

No target column was specified. The system applies Unsupervised Learning – discovering natural groupings in the data without predefined labels. Useful for operational mode detection and anomaly profiling. | Stratified 80/20 train-test split applied (stratify=y for classification). No data leakage – preprocessing fit exclusively on training fold.

Numerical missing values filled using median imputation. Categorical gaps filled using most-frequent (mode) strategy. Categorical columns ['skill_abr'] were one-hot encoded. All features standardised to zero mean and unit variance (StandardScaler). Final feature matrix contains 35 columns after encoding. | StandardScaler fitted on X_train only; transform applied to both train/test to prevent test-set leakage. Boolean dummy columns cast to int8 for memory efficiency.

[1] KMeans

- Fast, scalable centroid-based clustering. Ideal for partitioning machine operational modes (e.g., normal, degraded, critical) from unlabeled sensor telemetry.

[2] DBSCAN

→ Density-based algorithm that discovers clusters of arbitrary shape and automatically flags outliers as noise. Perfect for anomaly detection in

manufacturing process data where defects are rare but critical.

[illegible]

Silhouette Score : 0.9949

N Clusters : 35

N Noise Points : 21

AI MODEL INSIGHT

After training and comparing all recommended models, DBSCAN achieved the best performance. It formed 35 coherent clusters with a Silhouette Score of 0.9949 (scores near +1 indicate well-separated, compact clusters). | Model selection criterion: weighted F1 (classification) or RMSE (regression). All models trained with default hyperparameters; recommend GridSearchCV / Optuna for production hyperparameter optimisation.

■■ FEATURE INSIGHT

Feature importance scores are not directly extractable for this model type. Consider using SHAP for model-agnostic explanations. | Importances sourced from estimator's native feature_importances_ (tree-based) or mean(|coef|) (linear). For model-agnostic explanations, integrate SHAP TreeExplainer / KernelExplainer.

BIG DATA TELEMETRY & SCALE

Throughput Speed : 48.5 MB/s (213,768 records)

PySpark RAM Req. : 512.0 MB (2 executors)

Partition Column : 'skill_abr' (5 partitions)

Storage Format : Delta Lake / Apache Parquet (ACID compliant)

[illegible]

Compliance Rating: 75 / 100 (ISO 27001 Aligned)

EU AI Act Risk : Minimal Risk (Unsupervised Data Exploratory Analytics)

GDPR Privacy : Clean – No raw PII stored

Human Oversight : Mandatory (Human operator must approve critical fault actions)

→ Voluntary code of conduct and standard documentation.

ACTIONABLE RECOMMENDATIONS

1. Investigate extreme sensor spikes detected in columns ['job_id'] via IQR bounds.
2. Evaluate cluster cohesion (Silhouette Score: 0.9949) using PCA 2D projections.
3. Partition Delta Lake tables on column 'skill_abr' for sub-second distributed telemetry queries.
4. Deploy the saved model binary via the FastAPI /orchestrate endpoint for production inference.

End of ADSLM Report | Enterprise AI Evaluation